Data Science 500 Assignment

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Name and Surname: Shivaar Sewnarain

Student ITS No: 402308101

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QUESTION	EXAMINER MARKS	MODERATOR MARKS	REMARKS

ASSIGNMENT INSTRUCTIONS

Please tick each box to confirm completion.

Use Times New Roman font, size 12, with 1.5 line spacing throughout the document. Apply Harvard Referencing Style for all citations and references.

For essay-style assignments, please include the following sections:

Table of Contents Introduction

Main Body (with relevant subheadings) Conclusion

References

Submit the assignment in PDF format on Moodle. Use the specified cover page provided.

Include a signed declaration of originality.

DECLARATION OF ORIGINALITY:

I hereby declare that this assignment is my own work and has not been copied from any other source except where due acknowledgment is made. I affirm that all sources used have been properly cited and that this submission complies with the institution's policies on academic integrity and plagiarism.

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1. Introduction

A publishing analytics team needs to understand what influences book prices and ratings on Books to Scrape, a sandbox website built for web-scraping practice (Zyte n.d.). This project builds an end-to-end data science pipeline: obtain the data, clean it, explore it visually, and model it to answer a business question (Ozdemir 2016). Here the pipeline scrapes the catalogue, cleans the data, explores price and rating patterns, and fits a linear regression model that estimates book price from publicly available listing fields.

The work is implemented as a single Python script (main.py) run in PyCharm. Python was chosen because it is a general-purpose language with a large standard library and an extensive ecosystem of third-party packages (Das et al. 2024). The script is organised in four parts that follow the data science process: data acquisition, cleaning, visualisation, and modelling (Ozdemir 2016). The questions answered are:

1. What does the catalogue look like for price, rating, stock and genre?
2. Do star rating, stock level, title length or description length relate to price?
3. Can linear regression predict price from these features better than predicting the average price?

2. Methodology

2.1 Data acquisition

Data were obtained directly from Books to Scrape using the Python requests library (Reitz n.d.). Each downloaded page was parsed with BeautifulSoup using the lxml parser (Richardson n.d.). BeautifulSoup turns raw HTML into a navigable tree, so elements such as titles and prices can be located by their tag name and CSS class (Richardson n.d.).

From each listing card the script extracted: book title, price text (GBP), star rating as a word (One to Five), stock availability text, and the product URL. From each product detail page it extracted: genre (third breadcrumb item), full stock text, product description, and UPC. Together these fields cover the title, price, star rating and stock availability needed for the analysis.

All 50 catalogue pages were requested (20 books per page), giving the full catalogue of 1 000 books (Zyte n.d.). A 0.5 second delay was used between requests, following the common practice

of throttling a scraper so that it does not overload the target server (Mitchell 2018). The raw table was saved as books_raw.csv so the pipeline can be re-run without scraping again.

2.2 Data cleaning and preparation

PART B loads books_raw.csv and produces books_clean.csv using pandas for tabular cleaning and type conversion (McKinney 2022). Rows with missing values were dropped with pandas dropna (McKinney 2022). One row was removed because category or stock text was still missing after the detail-page step, leaving 999 books.

Cleaning steps applied in the script:

• Price: the £ symbol was removed and the value was stored as a float column price_gbp (McKinney 2022).

• Star rating: the words One–Five were mapped to integers 1–5 in rating_num using a Python dictionary, which stores key–value pairs for fast look-up (Das et al. 2024).

• Stock: text such as "In stock (22 available)" was parsed to an integer stock_qty; if no quantity appeared, stock_qty was set to 0.

• Category: whitespace was stripped; the labels "Default" and "Add a comment" were replaced with "Uncategorised".

• Description: missing descriptions were filled with an empty string, then lower-cased for length features (McKinney 2022).

• Derived features for modelling: title_length, title_words, description_length, description_words.

After cleaning, checks in the script confirmed: prices greater than 0, ratings between 1 and 5, non-negative stock quantities, and more than 100 books. The final modelling table has 999 rows and 13 columns. Dealing with missing values and converting fields to the correct data types are essential preparation steps before any analysis (Ozdemir 2016).

2.3 Data visualisation and exploration

PART C uses matplotlib for histograms, bar charts and scatter plots (Hunter 2007). Exploratory visualisation is used at this stage to reveal distributions and relationships before any model is built (Ozdemir 2016). Charts are saved in the charts folder. The exploration focuses on price patterns, ratings, genre averages, stock levels, and relationships between numeric features and price.

2.4 Predictive modelling

PART D fits a linear regression model to estimate price_gbp using scikit-learn (Pedregosa et al. 2011). Linear regression models a numeric response as an intercept plus a weighted sum of the input features (James et al. 2021). The feature set is limited to numeric fields produced in cleaning: rating_num, stock_qty, title_length and description_length. These are the listing attributes available after scraping; they were chosen because they are fully numeric, present for every clean row, and can be justified without inventing external data.

An 80/20 train–test split was applied with train_test_split (test_size=0.20, random_state=42), giving 799 training books and 200 test books (Pedregosa et al. 2011). The model was fitted on the training set only. Holding back a test set gives an honest estimate of how the model performs on unseen data (James et al. 2021). Performance was measured with mean absolute error (MAE) in pounds on the test set, and compared with a baseline that always predicts the training-set mean price (James et al. 2021). Genre was explored in visualisation but not one-hot encoded in this baseline model so the coefficients stay simple to interpret.

3. Results and key insights

3.1 Catalogue price landscape

Across the 999 clean books, prices range from £10.00 to £59.99. The mean price is £35.08. The histogram (Figure 1) shows prices spread across that range rather than clustering at a single value.

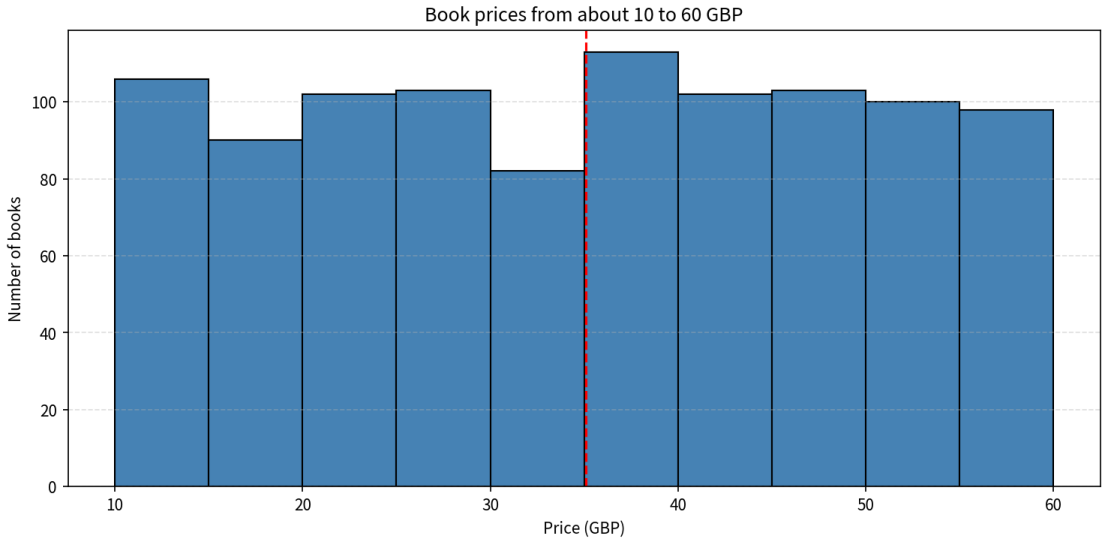


Figure 1. Distribution of book prices (GBP)

3.2 Ratings and price

Star ratings are fairly evenly distributed (179–225 books at each star level from 1 to 5). Average price by star rating stays close to the overall mean (Figure 2). There is no clear pattern that higher-rated books cost more on this catalogue.

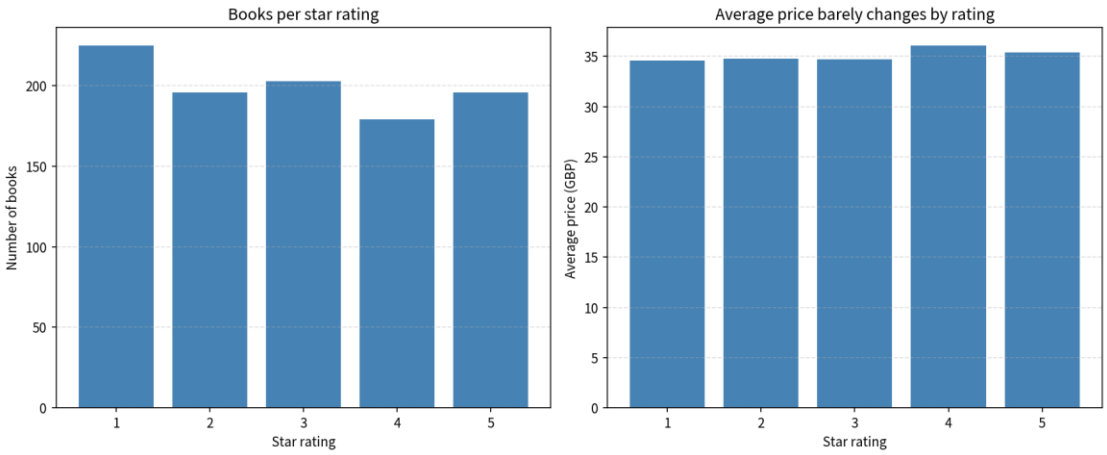


Figure 2. Number of books and average price by star rating

3.3 Genre and price

For genres with at least 20 books, Fantasy has the highest average price (about £39.59, 48 books). Food and Drink and Mystery sit lower (about £31–£32). The difference between the highest and lowest genre averages is only about £8, while single-book prices still run from £10 to £60. Genre therefore shows a small group-level difference but does not determine the price of an individual title. 218 books (about 22%) are labelled Uncategorised after cleaning messy site labels.

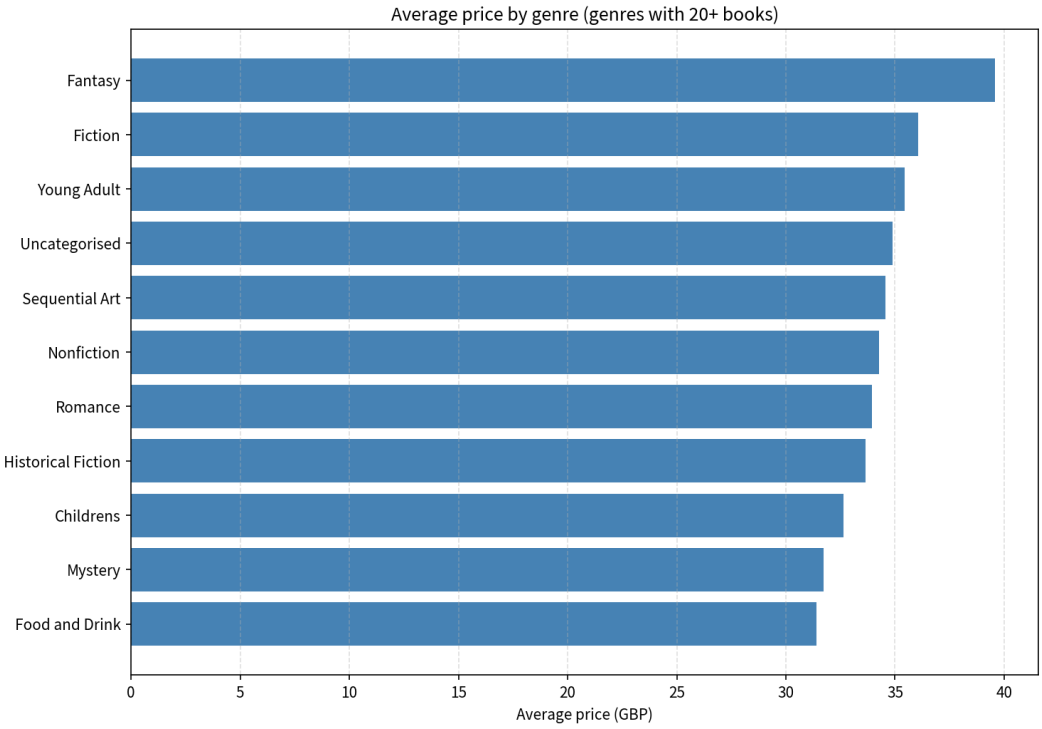


Figure 3. Average price by genre (genres with 20 or more books)

3.4 Stock levels

Stock quantities are not evenly spread. They bunch at specific values, especially 3 copies (196 books) and 14 copies (136 books) (Figure 4). Books to Scrape is a practice website built for scraping exercises (Zyte n.d.). This clustering is therefore more likely a fixed catalogue pattern than real restocking behaviour.

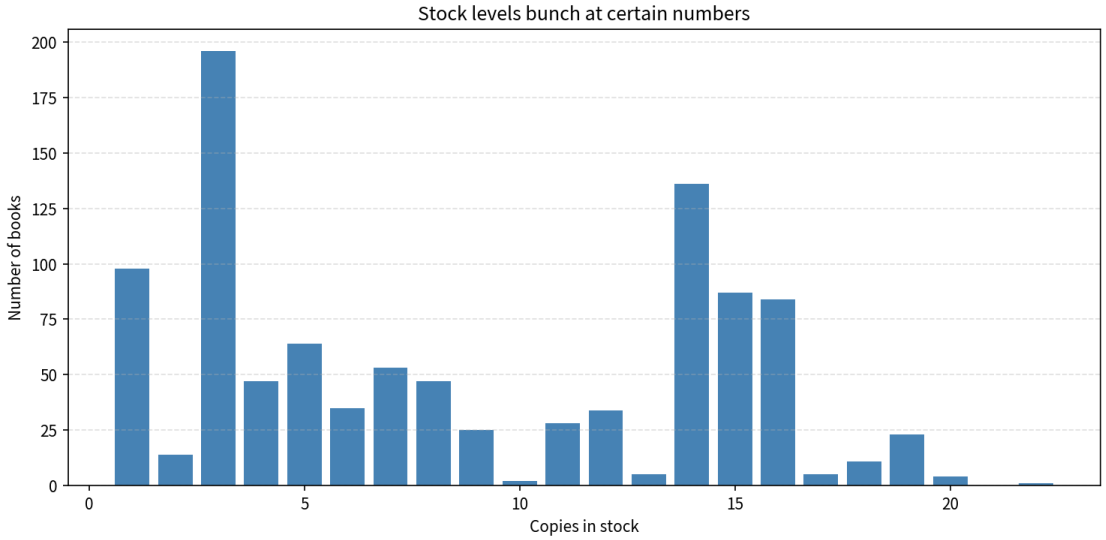


Figure 4. Distribution of copies in stock

3.5 Relationships between features and price

Scatter plots of price against rating_num, stock_qty, title_length and description_length are nearly flat (Figure 5). Pearson correlations with price_gbp, calculated with pandas, are all close to zero: rating_num 0.027, stock_qty −0.011, title_length 0.007, description_length 0.022 (Figure 6) (McKinney 2022). None of these listing fields shows a useful linear association with price.

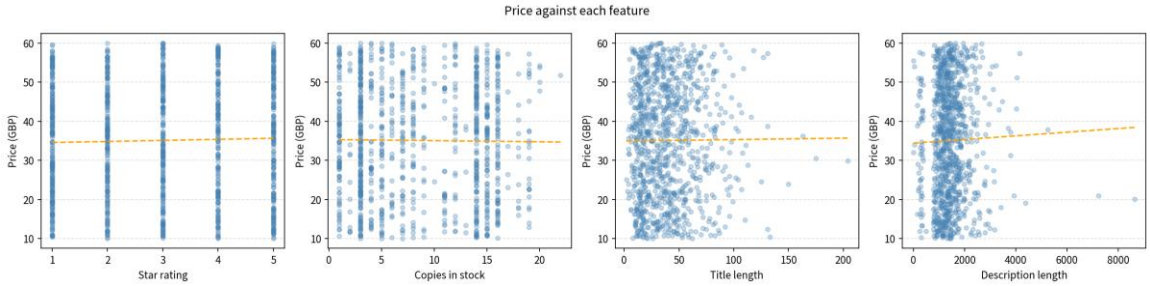


Figure 5. Price against each numeric feature (with fitted trend lines)

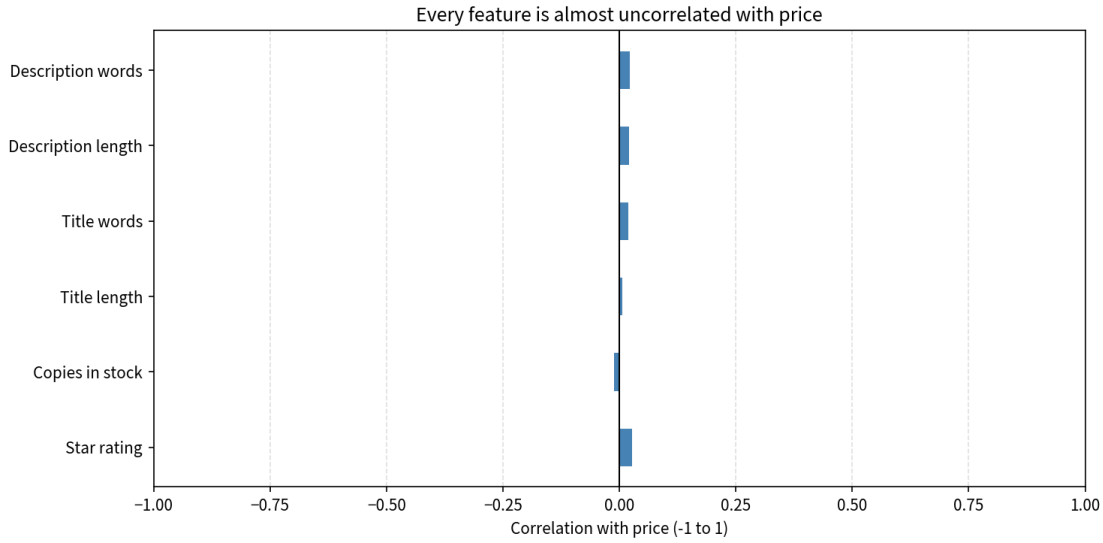


Figure 6. Correlation of each feature with price

4. Modelling outcomes

The fitted intercept is approximately 33.59. Estimated coefficients from the scikit-learn LinearRegression model are: rating_num ≈ 0.28, stock_qty ≈ −0.017, title_length ≈ 0.020, description_length ≈ 0.0003 (Pedregosa et al. 2011). Each coefficient is the expected change in

price for a one-unit change in that feature while the other inputs are held fixed (James et al. 2021). Moving from a 1-star to a 5-star rating therefore changes the predicted price by only about £1.12. That matches the weak correlations found in exploration.

On the 200 test books, mean absolute error for always predicting the training mean (£35.48) is £12.56. MAE for the linear regression model is £12.62. The regression does not outperform the simple average baseline. Figure 7 shows predicted prices sitting in a narrow band while actual prices still span roughly £10 to £60. Figure 8 compares the two average errors.

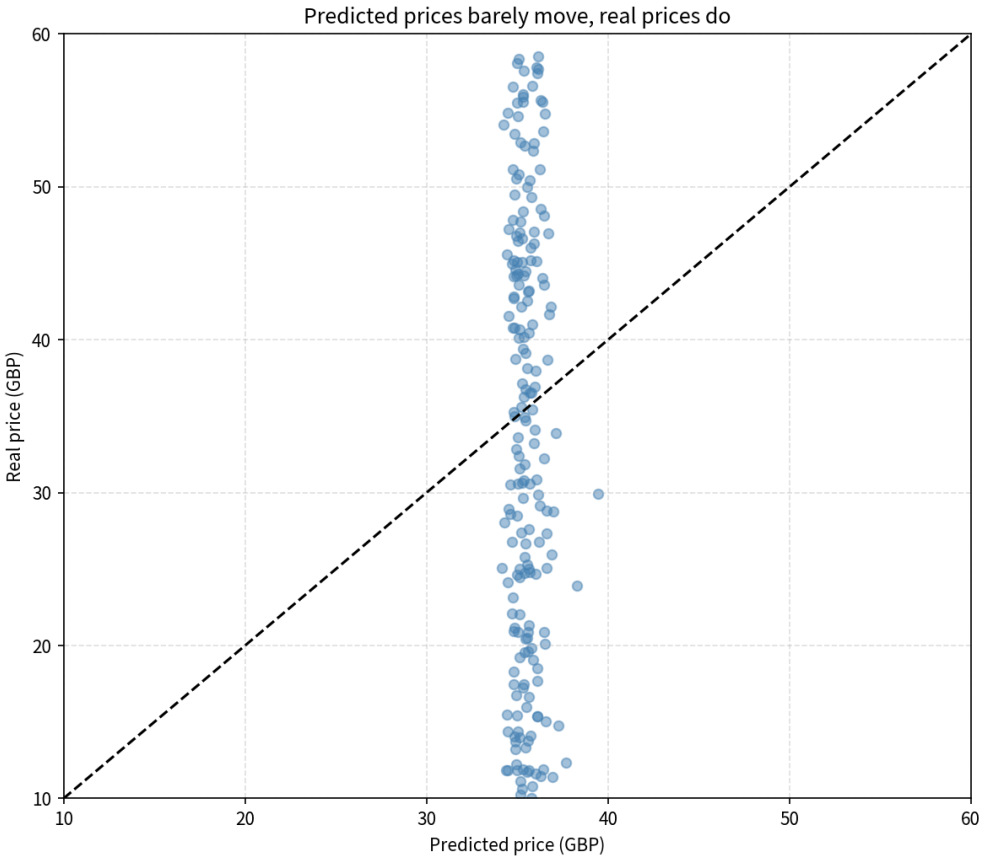


Figure 7. Actual test-set prices versus model predictions

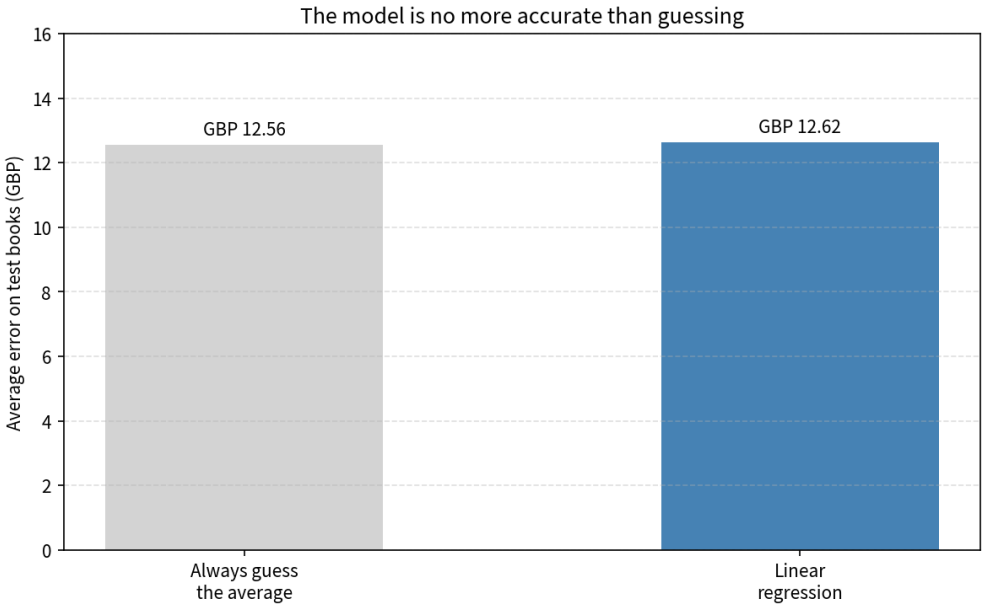


Figure 8. Average test error: mean baseline versus linear regression

This outcome is consistent with the data: if the features are almost uncorrelated with price, a linear model cannot recover a strong signal (James et al. 2021). The model is still correctly implemented and evaluated on a held-out test set against a simple baseline (James et al. 2021). The result shows that these public listing fields are not sufficient to estimate price on this catalogue.

5. Practical implications and recommendations

Retailers and online sellers

Do not use star rating, stock quantity, or title/description length alone to set prices for this type of catalogue. In this dataset those fields did not beat predicting the average price.

Publishers

Higher price does not line up with higher star rating on this site, and genre only shifts average price by a few pounds. Genre averages should not be treated as firm price rules.

Analytics team – next steps

Collect richer attributes before another pricing model (for example publisher, format, page count, or sales). Reduce Uncategorised genre labels. If trends matter, scrape the site again at later dates.

Validate any pricing model on real retailer data before operational use, because Books to Scrape is a practice website (Zyte n.d.).

6. Limitations

Books to Scrape is a sandbox site; its prices are not set by a real retailer (Zyte n.d.). The dataset is one snapshot with no time stamps, so change over time cannot be studied. The regression uses four numeric features only; genre was not encoded in the model. Linear regression only captures straight-line relationships between the inputs and the target (James et al. 2021). No sales or demand variable was available, so popularity cannot be modelled from this scrape alone.

7. Conclusion

The project completed the full pipeline: BeautifulSoup acquisition of 1 000 books, cleaning to a structured 999-row dataset, exploratory visualisations of price, rating, genre and stock, and a linear regression model for price. The main finding is that the listing features available on Books to Scrape (rating, stock, title length, description length) do not predict price better than the catalogue mean. That is a useful negative result for the analytics team: pricing decisions need richer data than these public listing fields provide.

8. References

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